Name	Index Number	Form Class	Tutorial Class	Subject Tutor

ANGLO-CHINESE JUNIOR COLLEGE
DEPARTMENT OF CHEMISTRY
Promotional Examination

CHEMISTRY **9729**
Higher 2

Answer Booklet 1 October 2018
2 hours

Additional Materials: Question Booklet
Data Booklet
Optical Answer Sheet (OAS)
Writing Papers

READ THESE INSTRUCTIONS CAREFULLY

Write your name and class on all the work you hand in. Write in dark blue or black pen. You may use a pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluids.

You are reminded of the need for good English and clear presentation in your answers. The number of marks is given in brackets [ ] at the end of each question or part question.

Shade your answers to Section A in the Optical Answer Sheet (OAS) provided. **No extra time will be given for the shading of answers.**

Answer **all** questions of Section B in the Answer Booklet.

Answer **all** questions of Section C on writing paper. Fasten your answers to Section C to this booklet.

For Examiner's Use		
	Question No.	**Marks**
Section A (Total: 15 m)		
Section B (Total: 35 m)	**1**	
	2	
	3	
Section C (Total: 25 m)	**1**	
	2	
	3	
	4	
TOTAL: 75 m		

This document consists of **9** printed pages, including this cover page.

Section B (35 marks)
Answer **all** questions in the spaces provided.

For Examiner's Use

1 Uranium, U, is an important element in nuclear research.

(a) Uranium-235 (^{235}U) is an <u>isotope of</u> <u>uranium</u> making up less than one percent of <u>natural uranium</u>.

(i) State the number of neutrons and protons in a ^{235}U atom. [1]

(ii) All the isotopes of uranium are unstable. [1]

When a ^{235}U atom is bombarded with a neutron, three neutrons, an atom of krypton-89 and a particle, **Y** are formed.

Y contains 88 neutrons and other particles.

Schematically,

$$^{1}_{0}\mathbf{n} + {}^{235}U \rightarrow 3\ {}^{1}_{0}\mathbf{n} + \mathbf{Y} + {}^{89}Kr$$

Determine the identity of **Y**.

(b) The most stable isotope of uranium is uranium-238.

For **(b)**, you are to use 238.0 as the relative atomic mass of uranium.

Excess water was added to 4.267 g of UF_6.

The only products were 3.730 g of a solid **Z** containing only uranium, oxygen and fluorine and 0.970 g of HF gas. The molecular formula of **Z** contains one U atom.

(i) Calculate the number of moles of HF and UF_6. [2]

For Examiner's Use

(ii) Hence, state the number of moles of **Z** formed. [1]

(iii) Using your answer in **b(ii)**, determine the molar mass of **Z**. [1]

(iv) Using your answers in **(b)**, determine the molecular formula of **Z**. [3]

[Total: 9]

2 Disulfur decafluoride (S_2F_{10}) is an extremely toxic gas which almost became a chemical weapon during World War II.

It exists in equilibrium with sulfur tetrafluoride (SF_4) and sulfur hexafluoride (SF_6).

$$S_2F_{10}\,(g) \rightleftharpoons SF_4\,(g) + SF_6\,(g)$$

(a) There is a S-S single bond in disulfur decafluoride. All the ten fluorine atoms are in equivalent environments in disulfur decafluoride.

(i) State the shape with respect to each sulfur in S_2F_{10}. [1]

For Examiner's Use

(ii) The boiling points of the three compounds are tabulated below. [2]

Compound	SF_4	SF_6	S_2F_{10}
Boiling point / K	235	209	303

Explain why SF_6 has a lower boiling point than SF_4.

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(iii) Explain why SF_4 has a lower boiling point than S_2F_{10}. [1]

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(b) This part of the question focuses on the thermodynamic aspects of the reaction.

$$S_2F_{10}\,(g) \rightleftharpoons SF_4\,(g) + SF_6\,(g)$$

(i) Using suitable data from the Data Booklet, determine the enthalpy change of the reaction. [2]

State the most important assumption you made.

For Examiner's Use

(ii) Explain why this decomposition becomes more thermodynamically feasible at higher temperatures. [2]

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(c) This part of the question focuses on the chemical equilibrium aspects of the reaction.

$$S_2F_{10}\,(g) \rightleftharpoons SF_4\,(g) + SF_6\,(g)$$

In a study of this decomposition, 2 moles of disulfur decafluoride was placed in a 2.00 dm^3 flask and heated up to 100 °C; its concentration became 0.50 mol dm^{-3} at equilibrium.

(i) Calculate the value of K_c at 100 °C. Include the units. [2]

For Examiner's Use

To this equilibrium mixture, a certain mass of disulfur decafluoride was added. The new equilibrium concentration of it became 2.50 mol dm^{-3}.

(ii) Calculate the mass of disulfur decafluoride added. [3]

(iii) Predict, with reasoning, how the *position of equilibrium* would change when the volume of the container is reduced. [1]

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[Total: 14]

3 This question is about interhalogens, compounds that are made of more than one type of halogen.

Chlorine trifluoride, ClF_3, is one of the most reactive chemicals ever known to man. It causes sand to explode.

(a) ClF_3 reacts with silver chloride according to the **un**balanced equation below.

$$2AgCl\ (s) + ClF_3\ (l) \rightarrow AgF_2\ (s) + 2ClF\ (g) + Cl_2\ (g)$$

(i) Balance the above equation by inserting the respective stoichiometric coefficients in the spaces provided below. Do fill in the coefficient even if it happens to be 1. [1]

$$2AgCl\ (s) + \textbf{......}ClF_3\ (l) \rightarrow \textbf{......}AgF_2\ (s) + 2ClF\ (g) + \textbf{......}Cl_2\ (g)$$

(ii) State the electronic configuration of the outermost shell of Ag in Ag*Cl*. [1]

(b) ClF_3 and $ClBr_3$ share the same molecular shape. However, the F-C*l*-F bond angle is predicted to be smaller than the Br-C*l*-Br angle.

(i) Explain why bromine is less electronegative than fluorine. [1]

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(ii) Hence, or otherwise, explain the relative sizes of the two bond angles. [2]

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For Examiner's Use

(c) Iodine forms many compounds with fluorine – IF, IF_3, IF_5 and IF_7.

The four compounds can interconvert into one another according to the following chemical equations.

(1) $3IF\,(s) \rightarrow I_2\,(s) + IF_3\,(s)$ $\quad \Delta H_1 = -199.8$ kJ mol^{-1}

(2) $7IF_5\,(l) \rightarrow I_2\,(s) + 5IF_7\,(g)$ $\quad \Delta H_2 = +1089$ kJ mol^{-1}

(3) $5IF_3\,(s) \rightarrow I_2\,(s) + 3IF_5\,(l)$ $\quad \Delta H_3$

(i) All three reactions can be classified under a certain type of redox reaction. Identify this type of reaction. [1]

The standard enthalpy changes of formation of IF and IF_7 are -95.4 and -962.5 kJ mol^{-1} respectively.

(ii) Calculate the standard enthalpy changes of formation of IF_3 and that of IF_5. [2]

(iii) Hence, calculate the standard enthalpy change of reaction **(3)**. [1]

(iv) Briefly explain why there are negligible changes in entropy in reactions **(1)** and **(3)**. [1]

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(v) Hence, predict whether reaction **(1)** or **(3)** has a lower likelihood of happening. [2]

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[Total: 12]

~ END OF SECTION B ~